IMAGE & VIDEO PROCESSING

Multi-Feature Fusion Based Cross-Camera Person Tracking System

30-Slide Presentation Content Plan

IVP Group 17

Indian Institute of Information Technology Allahabad
Department of Information Technology
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Prepared as a slide-by-slide guide for creating the project presentation.

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How to use this document

This document is a content guide, not the final PowerPoint. Use the headings and bullets to build the slides yourselves. Keep slide text concise, use diagrams where suggested, and explain the technical details orally. The project reports describe a proposed system, so do not claim that implementation is complete or that particular accuracy has been achieved unless your group has measured it. Results quoted from published papers must be clearly attributed to those papers.

Slide 1: Title Slide

Purpose: Introduce the project, course, institution, and team.

Include

- Project title: *Multi-Feature Fusion Based Cross-Camera Person Tracking System.*
- Image & Video Processing; Department of Information Technology; IIIT Allahabad.
- IVP Group 17; Academic Year 2026–2027.
- All seven team members and roll numbers, as listed on the cover page of this document.

Suggested visual

A surveillance-camera illustration showing the same person in three camera views. Keep the title slide uncluttered.

Slide 2: Presentation Roadmap

Purpose: Show the sequence of the presentation.

1. Introduction and problem definition.
2. Base research paper and literature review.
3. Research gap and proposed contribution.
4. System architecture and methodology.
5. Implementation and experimental evaluation.
6. Project planning, limitations, and conclusion.

Suggested visual

A six-part roadmap or horizontal progress line.

Slide 3: Introduction to Cross-Camera Person Tracking

Purpose: Define the task and distinguish its two main parts.

- Cross-camera person tracking attempts to identify the same individual across multiple camera views.
- Single-camera tracking maintains a person's identity across frames within one camera.
- Cross-camera association determines whether observations from different cameras belong to the same person.
- The goal is a consistent global identity across the camera network.

Example: Camera 1 observes a person; the person leaves its field of view; Camera 2 later observes the person; the system attempts to associate both observations with Global ID 1.

Suggested visual

Three camera panels with the same illustrative person marked Global ID 1.

Slide 4: Motivation and Real-World Applications

Purpose: Explain why the problem matters.

- Surveillance and public safety: connect observations across monitored locations.
- Smart campuses: analyse movement across campus locations.
- Transportation: study pedestrian movement in stations and terminals.
- Retail analytics: understand movement between store sections.
- Crowd analysis: study pedestrian trajectories and movement patterns.

Takeaway: Analysing cameras independently provides only a partial view of a person's movement. Cross-camera tracking attempts to connect observations into a consistent trajectory.

Suggested visual

Five simple application icons.

Slide 5: Problem Statement

Purpose: State the problem, inputs, processing, and outputs.

Problem statement: Design and develop a Multi-Feature Fusion Based Cross-Camera Person Tracking System that detects and tracks people within individual cameras and associates the same individuals across multiple camera views using appearance, motion, trajectory, and temporal information.

- **Input:** Recorded surveillance videos or supported live camera streams.
- **Processing:** Person detection, single-camera tracking, feature extraction, fusion, and cross-camera association.
- **Output:** Bounding boxes, local IDs, global IDs, trajectories, and visualisations.
- **Goal:** Reduce identity switches under occlusion, viewpoint variation, illumination changes, and re-entry.

Slide 6: Objectives of the Project

Purpose: List the project's concrete objectives.

1. Process video streams from multiple cameras.
2. Detect people in individual frames.
3. Maintain local identities across consecutive frames.
4. Extract appearance representations.
5. Analyse motion, trajectories, and temporal information.
6. Combine complementary features.
7. Associate observations across cameras.
8. Maintain global identities and evaluate tracking performance.

Slide 7: Base Research Paper

Purpose: Identify the paper that provides the conceptual foundation.

Title: "Cross-Camera Tracking Model and Method Based on Multi-Feature Fusion"
Authors: P. Zhang, S. Wang, W. Zhang, W. Lei, X. Zhao, Q. Jing, and M. Liu.
Publication: *Symmetry*, vol. 15, no. 12, Article 2145, 2023.
DOI: https://doi.org/10.3390/sym15122145

- The paper motivates combining complementary features for cross-camera association.
- The project's central adopted idea is that one feature alone may be unreliable under changing views and conditions.
- Our proposed work uses this concept as a foundation for a modular person-tracking pipeline.

Important note

Describe this as the conceptual basis of the project. Do not claim that the implementation reproduces every detail of the paper unless you verify and implement those details.

Slide 8: Understanding the Base Paper

Purpose: Explain the central concept and the proposed fusion representation.

- Obtain person observations from video.
- Extract appearance-related and contextual information.
- Combine complementary features.
- Compare observations across cameras and associate likely matches.

A conceptual fusion representation in the project design is:

$$F_{\text{fusion}} = w_a F_{\text{appearance}} + w_m F_{\text{motion}} + w_t F_{\text{temporal}}$$

Here, $F_{\text{appearance}}$, F_{motion}, and F_{temporal} represent different information sources; w_a, w_m, and w_t are configurable weights.

Important note

This is the project's conceptual design equation. Verify the original paper's exact feature definitions and implementation before claiming that it reproduces the paper exactly.

Slide 9: Why Single-Feature Matching Is Insufficient

Purpose: Explain the limitations of relying on only one type of information.

Feature	Information	Potential limitation
Appearance	Clothing, colour, texture, embeddings	Sensitive to lighting, pose, and viewpoint
Motion	Direction and movement	Different people can move similarly
Trajectory	Sequence of positions	Can be unreliable under occlusion or different camera geometry
Temporal	Observation times and elapsed time	Timing alone cannot uniquely identify a person

Conclusion: Complementary information may provide a more robust basis for matching, but fusion does not eliminate every failure mode.

Slide 10: Literature Survey: Existing Approaches

Purpose: Summarise the research directions reviewed by the group.

Direction	Example	Main idea
Multi-feature fusion	Zhang et al. (2023)	Combine complementary features
Graph-based tracking	ReST (2023)	Spatial-temporal graph modelling
Dynamic graph association	DyGLIP (2021)	Link prediction for association
Anchor-guided clustering	Huang et al. (2023)	Clustering and identity reassignment
Appearance + spatial-temporal fusion	Yang et al. (2023)	Combine visual and temporal cues
Geometry-based optimisation	LMGP (2022)	Geometry projections and lifted-multicut optimisation
Trajectory prediction	Paper reviewed in the group's survey	Use predicted movement in association

Main observation: Research explores fusion, graph models, clustering, geometry, and trajectory prediction to improve cross-camera identity consistency. Include full citations in the reference section.

Slide 11: Research Gap

Purpose: Explain the motivation for the proposed implementation.

- **Fusion weights:** Manually configured weights may not work equally well across scenes.
- **Identity switches and re-entry:** Occlusion and fragmented tracks can cause incorrect identity assignments.
- **Implementation complexity:** Advanced graph and geometry methods may require additional engineering and computation.
- **Accuracy versus speed:** High accuracy and fast multi-stream processing can be competing goals.

Project research direction: Build a modular, experimentally testable system with configurable feature fusion and spatio-temporal consistency checks. Present this as the motivation identified in the reviewed literature, not as a claim that no other work has addressed these issues.

Slide 12: Our Proposed Contribution

Purpose: Separate the proposed additions from existing methods.

1. **Configurable fusion:** Test different weights and feature combinations.
2. **Consistency checking:** Add a post-association check inspired by spatio-temporal identity reassignment research.
3. **Modular implementation:** Keep detection, tracking, extraction, fusion, association, and identity management separate.
4. **Experimental comparison:** Compare configurations with identity-based metrics.

Expected contribution: An experimental study of whether configurable fusion and consistency checks improve identity continuity. Report improvements only after experiments.

Slide 13: Proposed System Workflow

Purpose: Show the complete processing pipeline.

1. Multiple video streams.
2. Frame acquisition and preprocessing.
3. Person detection.
4. Single-camera multi-person tracking.
5. Appearance and motion feature extraction.
6. Multi-feature fusion.
7. Cross-camera association.
8. Spatio-temporal consistency checking.
9. Global identity management.
10. Visualisation and result storage.

Suggested visual

A clear flowchart with arrows. Use consistent boxes for processing modules.

Slide 14: System Architecture

Purpose: Group modules into three logical layers.

Input and Perception Layer

- Video input and preprocessing; person detection; single-camera tracking.

Feature and Association Layer

- Appearance extraction; motion and trajectory analysis; feature fusion; cross-camera association; consistency checking.

Identity and Output Layer

- Global identity management; visualisation; result storage and evaluation.

Design principle: Individual components can be tested or replaced without redesigning the complete pipeline.

Suggested visual

Three large layer boxes connected vertically.

Slide 15: Module 1: Video Input and Preprocessing

Purpose: Explain how frames enter the system.

1. Load recorded videos or supported live streams.
2. Read frames using OpenCV.
3. Resize frames if required by the chosen models.
4. Apply necessary preprocessing.
5. Preserve camera IDs and frame timestamps.
6. Pass frames to person detection.

Tools: Python, OpenCV, NumPy.
Output: Frames associated with camera IDs and timestamps. Recorded videos are a practical starting point before live-stream support.

Slide 16: Module 2: Person Detection

Purpose: Explain how people are located in each frame.

1. Supply a frame to the detector.
2. Generate candidate person detections.
3. Return bounding boxes and confidence scores.
4. Apply a configurable confidence threshold if needed.
5. Pass detections to the tracker.

Represent a detection as:

$$D_i = (x_i, y_i, w_i, h_i, c_i)$$

where (x_i, y_i) denotes position, w_i, h_i are width and height, and c_i is confidence.
Proposed option: A compatible pretrained YOLO-family detector.
Output: Person detections for each frame.

Suggested visual

A video frame with illustrative bounding boxes.

Slide 17: Module 3: Single-Camera Multi-Person Tracking

Purpose: Explain how local identities persist across frames.

1. Receive detections for the current frame.
2. Associate detections with existing tracks.
3. Create local IDs for unmatched detections.
4. Update positions and trajectory histories.
5. Handle missing detections according to tracker settings.

Camera	Frame	Local ID
Camera 1	10	C1-T1
Camera 1	11	C1-T1
Camera 1	12	C1-T1
Camera 1	13, another person	C1-T2

Proposed options: ByteTrack or DeepSORT, depending on compatibility and performance. A local ID is not necessarily a global ID.

Slide 18: Module 4: Appearance Feature Extraction

Purpose: Explain how visual characteristics are represented.

Appearance information can include clothing colour, texture, patterns, body-region appearance, and deep visual embeddings.

1. Crop the person using the bounding box.
2. Resize and preprocess the crop.
3. Run a selected appearance-feature extractor.
4. Obtain a feature vector and associate it with the track.

$$F_a \in R^{d_a}$$

Here, d_a is the dimensionality of the selected representation.
Options: A pretrained person re-identification model or a simpler appearance baseline. Confirm model availability, dependencies, and compatibility before finalising the choice.

Slide 19: Module 5: Motion and Trajectory Analysis

Purpose: Explain how movement and time are represented.

Maintain each track's current and previous positions, displacement, estimated velocity, timestamps, and track history.

$$F_m = \{\Delta x, \Delta y, v_x, v_y, t\}$$

For positions observed at two times:

$$v_x = \frac{x_t - x_{t-1}}{t - t_{t-1}}, \qquad v_y = \frac{y_t - y_{t-1}}{t - t_{t-1}}$$

Limitation: Coordinates from different cameras are not automatically comparable. Interpret motion in camera-specific coordinates unless a suitable geometric mapping is available.
Output: Motion descriptors and track histories.

Slide 20: Module 6: Multi-Feature Fusion

Purpose: Explain how complementary evidence is combined.

A conceptual feature-level representation is:

$$F_{\text{fusion}} = w_a F_a + w_m F_m + w_t F_t$$

However, arbitrary feature vectors of different dimensions cannot safely be added. The implementation must use compatible representations or fuse scalar similarity scores.

$$S(i,j) = w_a S_a(i,j) + w_m S_m(i,j) + w_t S_t(i,j)$$

- S_a: appearance similarity.
- S_m: motion compatibility.
- S_t: temporal compatibility.
- w_a, w_m, w_t: configurable weights.

Planned experiments: Appearance-only baseline; appearance plus motion; appearance plus motion and temporal information. Use only the configurations that are successfully implemented.

Slide 21: Module 7: Cross-Camera Association

Purpose: Explain how candidate tracks are compared.

1. Obtain candidate tracks from different cameras.
2. Compare appearance representations.
3. Evaluate available motion and temporal compatibility.
4. Combine scores and apply a configurable threshold.
5. Accept, reject, or defer the proposed association.

$$S(i,j) = \alpha S_a(i,j) + \beta S_m(i,j) + \gamma S_t(i,j)$$

A candidate may be accepted when $S(i,j) \geq \tau$ and other constraints are satisfied.
Important: A threshold alone does not guarantee correct matching. The system must handle conflicting matches and avoid incompatible simultaneous assignments.

Slide 22: Module 8: Global Identity and Consistency Checking

Purpose: Explain how local tracks become global identities.

Camera	Local ID	Global ID
Camera 1	C1-T4	G1
Camera 2	C2-T7	G1
Camera 3	C3-T2	G1

The system will examine suspicious associations using available evidence:

- Observation times and temporal compatibility.
- Conflicts with known track history.
- Implausible simultaneous observations.
- Camera geometry or temporal constraints, if available.

A person may legitimately appear in multiple overlapping cameras simultaneously. Consistency rules must allow compatible observations.
Output: Global IDs, local-to-global mappings, and association history.

Slide 23: Implementation Plan and Software Modules

Purpose: Connect the proposed design to development tasks.

1. Establish a video-loading and detection baseline.
2. Add local tracking and track-history storage.
3. Implement appearance and motion extraction.
4. Implement configurable fusion and cross-camera matching.
5. Add global identity management and consistency checks.
6. Evaluate the pipeline and document failures.

Suggested files:

- `main.py`: coordinate the pipeline.
- `video_input.py`: read videos and frames.
- `detector.py`: person detection.
- `tracker.py`: local tracking.
- `feature_extractor.py`: appearance features.
- `motion_analysis.py`: motion and trajectory features.
- `feature_fusion.py`: combine scores.
- `cross_camera_association.py`: match tracks.
- `global_identity.py`: maintain global IDs.
- `visualization.py`: display results.
- `evaluation.py`: calculate metrics.

These are proposed file names, not a claim that the files already exist.

Slide 24: Technology Stack

Purpose: List proposed tools and their roles.

Component	Technology	Purpose
Language	Python	Main implementation
Video processing	OpenCV	Frame I/O and visualisation
Numerical processing	NumPy	Feature and trajectory calculations
Deep learning	PyTorch	Compatible pretrained models
Detection	YOLO-family detector	Bounding boxes
Local tracking	ByteTrack or DeepSORT	Local identities
Appearance	Person Re-ID model	Appearance representation
Analysis	Pandas, NumPy	Evaluation data
Plots	Matplotlib/OpenCV	Results and visualisation

Linux or Windows may be used. CPU-based prototyping is possible; a compatible GPU is recommended for demanding deep-learning workloads. Final choices depend on compatibility, available weights, and measured performance.

Slide 25: Dataset Selection and Preparation

Purpose: Explain candidate datasets and preparation.

NVIDIA AI City Challenge – Track 1

- Candidate dataset for surveillance-like multi-camera scenes.
- Potentially useful for cross-camera identity evaluation and comparison with relevant published work.

WildTrack

- Candidate dataset containing seven calibrated, synchronised, overlapping cameras in a crowded pedestrian scene.
- Useful for studying crowding and occlusion in overlapping views.

Preparation process:

1. Obtain the dataset and check its licence and usage conditions.
2. Identify camera videos and ground-truth annotations.
3. Convert sequences to the input format expected by the pipeline.
4. Preserve ground truth for evaluation.
5. Run the same procedure for each configuration.

Begin with a manageable subset of one dataset. Do not claim evaluation on a dataset until the experiments have been run. Overlapping-camera results do not by themselves establish performance on non-overlapping cameras.

Slide 26: Experimental Design and Evaluation Metrics

Purpose: Define how performance will be measured.

Detection: precision, recall, and bounding-box quality where ground truth is available.
Tracking and association: ID Precision (IDP), ID Recall (IDR), IDF1, and identity-switch count.
Computational performance: frames per second (FPS), runtime, and multi-stream performance where supported.

$$IDP = \frac{IDTP}{IDTP + IDFP}$$

$$IDR = \frac{IDTP}{IDTP + IDFN}$$

$$IDF1 = \frac{2IDTP}{2IDTP + IDFP + IDFN}$$

$IDTP$ denotes correctly matched identity detections, $IDFP$ incorrect identity detections, and $IDFN$ missed ground-truth identity detections. Higher IDF1 indicates a better balance of identity precision and recall. Use a consistent evaluation protocol.

Slide 27: Experimental Comparison and Ablation Study

Purpose: Show the experiments that will test the proposed additions.

Test	Configuration	Purpose
E1	Detection and local tracking	Establish the basic pipeline
E2	Appearance-based association	Appearance baseline
E3	Appearance + motion fusion	Measure motion contribution
E4	Appearance + motion + temporal information	Measure temporal contribution
E5	Selected fusion + consistency checks	Measure proposed enhancement

Record IDF1, IDP, IDR, identity switches, and FPS where applicable. Keep the dataset, sequences, and underlying pipeline consistent between comparisons.

Configuration	IDF1	ID switches	FPS
Appearance baseline	To be measured	To be measured	To be measured
Appearance + motion	To be measured	To be measured	To be measured
Fusion + consistency check	To be measured	To be measured	To be measured

Important note

Do not insert invented experimental values.

Slide 28: Expected Outcomes, Limitations, and Future Work

Purpose: Summarise intended deliverables and constraints.

Expected outcomes

- Person detection and local tracking pipeline.
- Appearance and motion feature extraction.
- Configurable fusion and cross-camera association.
- Global identity management and visualisation.
- Quantitative comparison of implemented configurations.

Limitations

- Occlusion and missed detections can fragment tracks.
- Similar clothing and viewpoint changes can confuse appearance matching.
- Camera-coordinate differences complicate motion comparison.
- Multi-stream processing increases computational demand.

Future work

- Learn fusion weights automatically.
- Explore graph-based association.
- Improve recovery from occlusion and re-entry.
- Use camera calibration where appropriate.
- Optimise inference for multiple streams.

Clearly label outcomes as goals until demonstrated.

Slide 29: Project Timeline and Work Plan

Purpose: Show the remaining work and deliverables.

The following is a proposed schedule; adjust it to the actual course deadline and group progress.

Period	Activity	Deliverable
16–20 Oct	Finalise dataset and setup	Dataset and environment
21–27 Oct	Detection and local tracking	Working baseline
28 Oct–3 Nov	Feature extraction	Appearance/motion modules
4–10 Nov	Fusion implementation	Configurable fusion
11–17 Nov	Cross-camera association	Initial global IDs
18–24 Nov	Consistency and visualisation	Integrated pipeline
25 Nov–1 Dec	Experiments and metrics	Results and plots
2–8 Dec	Analysis and documentation	Final report draft
9–12 Dec	Presentation and demo	Final submission materials

Suggested visual

A Gantt chart with dates along the horizontal axis. Mark work complete only when a working deliverable exists.

Slide 30: References and Questions

Purpose: Close with citations and invite questions.

1. P. Zhang, S. Wang, W. Zhang, W. Lei, X. Zhao, Q. Jing, and M. Liu, "Cross-Camera Tracking Model and Method Based on Multi-Feature Fusion," *Symmetry*, vol. 15, no. 12, Art. 2145, 2023. doi: 10.3390/sym15122145.
2. C.-C. Cheng, M.-X. Qiu, C.-K. Chiang, and S.-H. Lai, "ReST: A Reconfigurable Spatial-Temporal Graph Model for Multi-Camera Multi-Object Tracking," in *Proc. IEEE/CVF ICCV*, 2023, pp. 10051–10060.
3. K. G. Quach et al., "DyGLIP: A Dynamic Graph Model With Link Prediction for Accurate Multi-Camera Multiple Object Tracking," in *Proc. IEEE/CVF CVPR*, 2021, pp. 13784–13793.
4. H.-W. Huang et al., "Enhancing Multi-Camera People Tracking With Anchor-Guided Clustering and Spatio-Temporal Consistency ID Re-Assignment," in *Proc. IEEE/CVF CVPR Workshops*, 2023, pp. 5239–5249.
5. Yang et al., "Integrating Appearance and Spatial-Temporal Information for Multi-Camera People Tracking," in *Proc. IEEE/CVF CVPR Workshops*, 2023, pp. 5260–5269.
6. Nguyen et al., "LMGP: Lifted Multicut Meets Geometry Projections for Multi-Camera Multi-Object Tracking," in *Proc. IEEE/CVF CVPR*, 2022, pp. 8856–8865.
7. Add the complete verified citation for the trajectory-prediction paper reviewed by the group.

Closing text: Thank You – Questions & Discussion.

Important note

Verify full author lists, exact titles, publication details, and the seventh paper against the original publications or the group's literature-survey workbook before submission. Split references over two slides if the text becomes too small.

Final checklist before building the PowerPoint

- Confirm the base paper's citation and explain how it informs the project.
- Clearly distinguish existing methods from the group's proposed work.
- Create original workflow and architecture diagrams.
- Check that the selected detector, tracker, and feature extractor are compatible.
- Use the same dataset and protocol for fair experimental comparisons.
- Include actual results only after running experiments.
- Update the timeline to reflect real progress and course deadlines.
- Verify all citations and make sure the slides remain readable.